

Due: Thursday, November 30th at the beginning of class.

Read Section 2.5 and 3.2 of *Levin*.

From *Levin*, complete the following:

Section 3.2: # 3, 5, 8, 16

Section 2.5: # 3, 6, 10

1. We say that a divides b , written $a|b$, if there is some integer k such that $ak = b$.
 - (a) Show that $3|12$ with the definition.
 - (b) Show that $3|9$ with the definition.
 - (c) What can you say about $3|(12 + 9)$, using the previous two answers.
 - (d) Show that $3|0$ with the definition.
 - (e) Prove that if $a|b$ and $a|c$, then $a|(b + c)$.
2. Prove that $\sqrt{7}$ is irrational. (You may assume that if n^2 is a multiple of 7, then n is a multiple of 7.)
3. Prove that $8 + 11 + 14 + \cdots + (3n + 5) = \frac{3n^2 + 13n}{2}$, for $n = 1, 2, 3, \dots$ via induction.
4. Prove that $3|4^n - 1$ for $n = 0, 1, 2, \dots$ via induction.

Solutions

Section 3.2

3. Solution is in Book.

5. On Homework 10.

8. Prove the statement: For all integers n , if $5n$ is odd, then n is odd. Clearly state the style of proof you are using.

Proof. We proceed by contrapositive. That means we will prove the statement *If n is not odd, then $5n$ is not odd.* As n is an integer, not odd is the same as even.

As n is even, there is some integer k such that $n = 2k$. Then $5n = 5(2k) = 2(5k)$. Let $\ell = 5k$. Then $5n = 2\ell$, which is the definition of an even integer. Thus, $5n$ is even, and therefore, if $5n$ is odd, then n is odd. $\square$

16. Your “friend” has shown you a “proof” he wrote to show that $1 = 3$. Here is the proof:

Proof. I claim that $1 = 3$. Of course we can do anything to one side of an equation as long as we also do it to the other side. So subtract 2 from both sides. This gives $-1 = 1$. Now square both sides, to get $1 = 1$.

And we all agree this is true. $\square$

What is going on here? Is your friend’s argument valid? Is the argument a proof of the claim $1 = 3$? Carefully explain using what we know about logic.

Solution

As I tell you all the time in class, if you want to show an equality is true, you must start with one side and get it to look like the other. But your “friend” is not doing that. They are starting with the equation they want to prove and manipulating it. AND THEY ARE ASSUMING IMPLICITLY THAT IT IS TRUE. Don’t be that guy.

Section 2.5

3. Solution is in Book.

6. Solution is in Book.

10. Prove that the sum of squares can be found as follows.

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof. Let $P(n)$ be the statement above. We will prove that $P(n)$ is true for $n = 1, 2, 3, \dots$ by induction.

Base Case: We will show that $P(1)$ is true. On the LHS, we have $1^2 = 1$. On the RHS, we have $\frac{1(1+1)(2(1)+1)}{6} = \frac{6}{6} = 1$. This shows the LHS = RHS and our base case is true.

Inductive Hypothesis: We will assume that $P(k)$ is true. That is

$$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

Inductive Step: Now we want to show that $P(k+1)$ is true, using $P(k)$. That is

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6}$$

We will operate on the LHS of $P(k+1)$ and show it is equal to the RHS. Rewriting the LHS, we have

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = (1^2 + 2^2 + 3^2 + \cdots + k^2) + (k+1)^2$$

On the right, we see that this contains the LHS of the inductive hypothesis, $P(k)$. So we substitute this out for the RHS of $P(k)$.

$$\begin{aligned} (1^2 + 2^2 + 3^2 + \cdots + k^2) + (k+1)^2 &= \left(\frac{k(k+1)(2k+1)}{6} \right) + (k+1)^2 \\ &= \frac{k(k+1)(2k+1)}{6} + \frac{6(k+1)^2}{6} \\ &= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} \end{aligned}$$

From here, we factor out a $(k+1)$.

$$\begin{aligned} \frac{k(k+1)(2k+1) + 6(k+1)^2}{6} &= \frac{(k+1)[k(2k+1) + 6(k+1)]}{6} \\ &= \frac{(k+1)[2k^2 + k + 6k + 6]}{6} \\ &= \frac{(k+1)[2k^2 + 7k + 6]}{6} \\ &= \frac{(k+1)(k+2)(2k+3)}{6} \\ &= \frac{(k+1)((k+1)+1)(2(k+1)+1)}{6} \end{aligned}$$

And this is the RHS of $P(k+1)$, which shows that our inductive step is true. Therefore, by induction, we have shown that $P(n)$ is true for all $n = 1, 2, 3, \dots$ $\square$

Problems from HW Sheet

1. We say that a divides b , written $a|b$, if there is some integer k such that $ak = b$.

(a) Show that $3|12$ with the definition.

Notice that $3(4) = 12$ and so 3 divides 12.

(b) Show that $3|9$ with the definition.

Notice that $3(3) = 9$ and so 3 divides 9.

(c) What can you say about $3|(12+9)$, using the previous two answers.

Notice that $12+9 = 3(4) + 3(3) = 3(4+3)$ and so 3 divides $(12+9)$.

(d) Show that $3|0$ with the definition.

Notice that $3(3) = 9$ and so 3 divides 9.

(e) Prove that if $a|b$ and $a|c$, then $a|(b+c)$.

Proof. We proceed directly. As $a|b$ and $a|c$, there are some integers k and j such that $ak = b$ and $aj = c$. Then, using part (c) as a guide, we see that $b+c = ak + aj = a(k+j)$. Thus, $a|(b+c)$. $\square$

2. Prove that $\sqrt{7}$ is irrational. (You may assume that if n^2 is a multiple of 7, then n is a multiple of 7.)

Proof. We begin by rewriting this as the statement *if $x = \sqrt{7}$, then x is irrational*. We proceed by contradiction. So we assume that $x = \sqrt{7}$ and x is not irrational. Thus x is rational, which means that x can be written as a fraction of integers p and q . We also can make the further assumption that p and q have no common factors. Therefore, we have

$$\begin{aligned}\sqrt{7} &= \frac{p}{q} \\ \left(\sqrt{7}\right)^2 &= \left(\frac{p}{q}\right)^2 \\ 7 &= \frac{p^2}{q^2} \\ 7q^2 &= p^2.\end{aligned}$$

This tells us that p^2 is a multiple of 7. By the assumption with the question, this tells us that p is a multiple of 7. And therefore, there is some integer k such that $p = 7k$. Squaring this equation, we get that $p^2 = 49k^2$. Then, we substitute this into the main equation we are working with and get

$$\begin{aligned}7q^2 &= p^2 \\ 7q^2 &= 49k^2 \\ q^2 &= 7k^2.\end{aligned}$$

Then we see that q^2 is a multiple of 7 and again, by the assumption, q is a multiple of 7. But we assumed at the beginning that p and q have no common factors and we just found that they both must be multiples of 7, which is a contradiction. Thus our assumption that x is rational must be false and therefore we have shown that if $x = \sqrt{7}$, then x is irrational. $\square$

3. Prove that $8 + 11 + 14 + \cdots + (3n + 5) = \frac{3n^2 + 13n}{2}$, for $n = 1, 2, 3, \dots$ via induction.

Done in class.

4. Prove that $3 \mid 4^n - 1$ for $n = 0, 1, 2, \dots$ via induction.

Done in class.